ARMCMC: ONLINE BAYESIAN DENSITY ESTIMATION OF MODEL PARAMETERS

Anonymous authors
Paper under double-blind review

ABSTRACT

Although the Bayesian paradigm provides a rigorous framework to estimate the full probability distribution over unknown parameters, its online implementation can be challenging due to heavy computational costs. This paper proposes Adaptive Recursive Markov Chain Monte Carlo (ARMCMC) which estimates full probability density of model parameters while alleviating shortcomings of conventional online approaches. These shortcomings include: being solely able to account for Gaussian noise, being applicable to systems with linear in the parameters (LIP) constraint, or having requirements on persistence excitation (PE). In ARMCMC, we propose a variable jump distribution, which depends on a temporal forgetting factor. This allows one to adjust the trade-off between exploitation and exploration, depending on whether there is an abrupt change to the parameter being estimated. We prove that ARMCMC requires fewer samples to achieve the same precision and reliability compared to conventional MCMC approaches. We demonstrate our approach on two challenging benchmarks: the estimation of parameters in a soft bending actuator and the Hunt-Crossley dynamic model. Our method shows at-least 70% improvement in parameter point estimation accuracy and approximately 55% reduction in tracking error of the value of interest compared to recursive least squares and conventional MCMC.

1 INTRODUCTION

Bayesian methods are powerful tools to not only obtain a numerical estimate of a parameter but also to give a measure of confidence (Kuśmierczyk et al., 2019; Bishop, 2006; Joho et al., 2013). By calculating the probability distribution of parameters rather than a point estimate, which is prevalent in frequentist paradigms (Tobar, 2018). One of the main advantages of probabilistic frameworks is that they enable decision making under uncertainty (Noormohammadi-Asl & Taghirad, 2019). In addition, knowledge fusion is significantly facilitated in probabilistic frameworks; different sources of data or observations can be combined according to their level of certainty in a principled manner (Agand & Shoorehdeli, 2019). Nonetheless, Bayesian inference requires high computational effort for obtaining the whole probability distribution and require prior general knowledge about the noise distribution.

One of the most effective methods for Bayesian inferences is Markov Chain Monte Carlo (MCMC). In the field of system identification, MCMC variants such as the one recently proposed by Green (2015) are mostly focused on offline system identification. This is partly due to computational challenges which prevent its real-time use (Kuindersma et al., 2012). The standard MCMC algorithm is not suitable for multi mode system since different candidates do not share the same parameter set. Green (1995) first introduced reversible jump Markov chain Monte Carlo (RJMCMC) as a method to address the model selection problem. In this method, an extra pseudo-random variable is defined to address dimension mismatch. There are further extensions of MCMC in the literature, however, there is a lack of variants suitable for online estimation.

Motion filtering and force prediction of robotic manipulators are important fields of study with interesting challenges suitable for Bayesian inference to address (Saar et al., 2018). Here, measurements are inherently noisy, which is not desirable for control purposes. Likewise, inaccuracy, inaccessibility, and costs are typical challenges that make force measurement not ideal for practical use (Agand et al., 2016). Different environmental identification methods have been proposed in the lit-

erature for linear and Gaussian noise (Wang et al., 2018); however, in cases of nonlinear models like Hunt-Crossley that does not have Gaussian noise (e.g. impulsive disturbance), there is no optimal solution for the identification problem. Diolaiti et al. (2005) proposed a double-stage bootstrapped method for online identification of the Hunt-Crossley model, which is sensitive to parameter initial conditions. Carvalho & Martins (2019) proposed a method to determine the damping term in the Hunt-Crossley model. A neural network-based approach was introduced to control the contact/non-contact Hunt-Crossley model by Bhasin et al. (2008)

This paper proposes a new technique, Adaptive Recursive Markov Chain Monte Carlo (ARMCMC), to address several weaknesses of traditional online identification methods, such as only being applicable to systems Linear in Parameters (LIP), having Persistent Excitation (PE) requirements, and assuming Gaussian noise. ARMCMC is an online method that takes advantage of the previous posterior distribution, whenever there is no sudden change in the parameter distribution. To achieve this, we define a new *variable jump distribution* that accounts for the degree of model mismatch using a *temporal forgetting factor*. The temporal forgetting factor is computed from a model mismatch index and determines whether ARMCMC employs modification or reinforcement to either restart or refine the estimated parameter distribution. As this factor is a function of the observed data rather than a simple user-defined constant, it can effectively adapt to the underlying dynamics of the system. We demonstrate our method using two different examples: a soft bending actuator and the Hunt-Crossley model. We show favorable performance compared to state-of-the-art baselines.

The rest of this paper is organized as follows: In Sec. 2, introductory context about the Bayesian approach and MCMC is presented. Sec. 3 is devoted to presenting the proposed ARMCMC approach and providing a step-by-step algorithm. Simulation results for the soft bending actuator and empirical results for a reality-based model of a soft contact environment capturing the Hunt-Crossley dynamic are presented in Sec. 4. Lastly, the final remarks and future directions are presented in Sec. 5.

2 PRELIMINARIES

2.1 PROBLEM STATEMENT

In the Bayesian paradigm, estimates of parameters are given in the form of the posterior probability density function (pdf); this pdf can be continuously updated as new data points are received. Consider the following general model:

$$Y = F(X, \theta) + \nu, \tag{1}$$

where Y, X, θ, and ν are concurrent output, input, model parameters and noise vector, respectively. To calculate the posterior pdf, the observed data (input/output pairs) along with a prior distribution are combined via Bayes' rule (Khatibisepehr et al., 2013). We will be applying updates to the posterior pdf using batches of data points; hence, it will be convenient to partition the data as follows:

$$D^t = \{(X, Y)_{t_m}, (X, Y)_{t_m+1}, \cdots, (X, Y)_{t_m+N_s+1}\}, \tag{2}$$

where $N_s = T_s/T$ is the number of data points in each data pack with T, T_s being the data and algorithm sampling times, respectively. This partitioning is convenient for online applications, as D^{t-1} should have been previously collected so that the algorithm can be executed from t_m to $t_m + N_s+1$ an interval which we will define as algorithm time step t. Ultimately, inferences are completed at $t_m + N_s + 2$. Fig. 1 illustrates the timeline for the data and the algorithm. It is worth mentioning that the computation can be done in parallel with the task done in the adjacent algorithm step (e.g. phase A of algorithm t, phase B of algorithm $t-1$ and phase C of algorithm $t-2$ can all be done simultaneously)

According to Bayes' rule and assuming data points are independent and identically distributed (i.i.d.) in Eq. (1), we have

$$P(\theta^t|[D^{t-1}, D^t]) = \frac{P\big(D^t|\theta^t, D^{t-1}\big)P(\theta^t|D^{t-1})}{\int P\big(D^1|\theta^t, D^{t-1}\big)P(\theta^t|D^{t-1})d\theta^t}, \tag{3}$$

where θ^t denotes the parameters at current algorithm time step. $P(\theta^t|D^{t-1})$ is the prior distribution over parameters, which is also the posterior distribution at the previous algorithm time step.

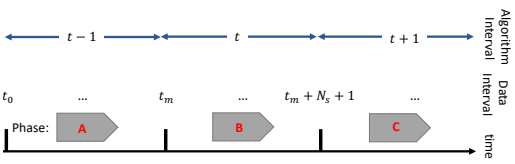


Figure 1: Data timeline and different phases of ARMCMC algorithm. For algorithm at time t: Phase (A) Data collection [we pack N_s data points for the next algorithm time step], Phase (B) Adjusting [the method is applying to the most recent data pack], (C) Execution [after the min evaluation of algorithm, the results will be updated on parameters posterior distribution and any byproduct value of interest].

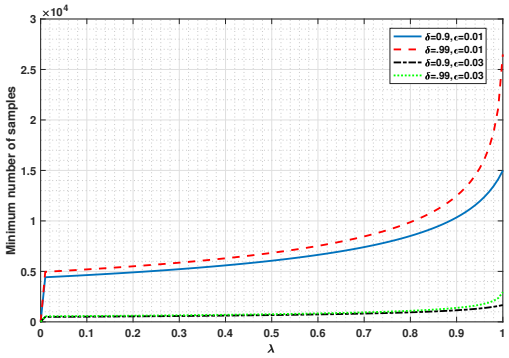


Figure 2: K_{min} with respect to λ for some values of ϵ, δ in ARMCMC. (for $\lambda = 1$ evaluation for ARMCMC is equivalent to MCMC)

$P\big(D^t|\theta^t, D^{t-1}\big)$ is the likelihood function which is obtained by sampling from the one-step-ahead prediction:

$$\hat{Y}^{t|t-1} = F(D^{t-1}, \theta^t), \tag{4}$$

where $\hat{Y}^{t|t-1}$ is a sample from the prediction of the output in (1). If the model in (4) is accurate, then the difference between the real output and predicted should be measurement noise, (i.e., $Y^{t|t-1} - \hat{Y}^{t|t-1} = \nu$). Therefore, the model parameter may be updated as follows:

$$P\big(D^t|\theta^t, D^{t-1}\big) = \prod_{t=t_m+1}^{t_m+N_s+1} P_\nu\big(Y^{t|t-1} - \hat{Y}^{t|t-1}\big), \tag{5}$$

where P_ν is the probability distribution of noise. Note that there is no restriction on the type of noise probability distribution.

Remark 1: As it was mentioned before, there is no need to know the exact probability distribution of noise. This probability distribution can be simply substituted with a Gaussian distribution, mean and variance of the data (Bishop, 2006).

2.2 Markov Chain Monte Carlo (MCMC)

MCMC is often employed to compute the posterior pdf numerically. The multidimensional integral in (3) is approximated by samples drawn from the posterior pdf. The samples are first drawn from a different distribution called proposal distribution, denoted $q(\cdot)$, which can be sampled more easily compared to the posterior. Brooks et al. (2011) discuss different types of MCMC implementations which may employ various proposal distributions and corresponding acceptance criteria. The main steps of the Metropolis-Hastings algorithm are listed as follows (Ninness & Henriksen, 2010):

1. Set initial guess θ_0 while $P(\theta_0|Y) > 0$ for iteration $k = 1$,
2. Draw candidate parameter θ_{cnd}, at iteration k, from the proposal distribution, $q(\theta_{cnd}|\theta_{k-1})$
3. Compute the acceptance probability,

$$\alpha(\theta_{cnd}|\theta_{k-1}) = \min\left\{1, \frac{P(\theta_{cnd}|D)q(\theta_{k-1}|\theta_{cnd})}{P(\theta_{k-1}|D)q(\theta_{cnd}|\theta_{k-1})}\right\}, \tag{6}$$

4. Generate a uniform random number γ in $[0, 1]$,
5. 'Accept' candidate if $\gamma \leq \alpha$ and 'ignore' it if $\gamma > \alpha$,
6. Set iteration to $k + 1$ and go to step 2.

2.3 PRECISION AND RELIABILITY

Two important notions in probabilistic frameworks to compare results are precision (ϵ) and reliability (δ). The former represents the proximity of a sample to the ground truth, and the latter represents the probability that an accepted sample lies within ϵ of the ground truth.

Lemma: Let P_k be k samples from MCMC, and $\mathbb{E}(P_k)$ denote their expected value. According to Chernoff bound by Tempo et al. (2012), to achieve the given precision and reliability given $\epsilon, \delta \in [0, 1]$, if the minimum number of samples (k) satisfies

$$k \geq \frac{1}{2\epsilon^2} \log(\frac{2}{1-\delta}), \tag{7}$$

then $Pr\Big\{\{P_k - \mathbb{E}(P_k)\} \leq \epsilon\Big\} \geq \delta$.

3 ARMCMC ALGORITHM

At each algorithm time interval, ARMCMC recursively estimates the posterior pdf by drawing samples. The number of samples drawn is constrained by the desired precision and reliability, and the real time requirement. On the other hand, the maximum number of data points in each data pack, N_s, is limited by the frequency of model variation, and the minimum is confined by the shortest required time such that the algorithm is real-time.

We propose a variable jump distribution that enables both exploiting and exploring. This will necessitate the definition of the temporal forgetting factor as a model mismatch measure to reflect current underlying dynamics of the data. We also prove that ARMCMC achieves the same precision and reliability with fewer samples compared to the traditional MCMC. Algorithm 1 summarizes ARMCMC.

3.1 VARIABLE JUMP DISTRIBUTION

We propose a variable jump distribution (also known as a proposal distribution) to achieve faster convergence, thereby enabling real-time parameter estimation:

$$q_k^t(\theta^t|\theta_{k-1}^t) = \begin{cases} P(\theta^{t-1}|D^{t-1}) & \lambda_k \leq \lambda^t \\ N(\mu_D, \sigma_\nu) & \lambda_k > \lambda^t \end{cases}, \tag{8}$$

where θ_{k-1}^t is the $(k-1)$-th parameter sample which is given by the t-th data pack throughout the MCMC evaluation. In each algorithm time sample, average of the second half of this quantity will construct θ^t. $P(\theta^{t-1}|D^{t-1})$ is the posterior pdf of the parameters at the previous algorithm time step, and $N(\mu_D, \sigma_\nu)$ is a Gaussian distribution with its mean and variance μ_D, σ_ν computed using the empirical mean and variance of D^{t-1}.

The hyperparameter λ^t (*temporal forgetting factor*), is an adaptive threshold for the t-th pack that takes inspiration from classical system identification; it regulates how previous knowledge affects the posterior pdf. A smaller value of λ^t intuitively means that there may be a large sudden change in ground truth value of θ, and thus more exploration is needed. Conversely, a larger value of λ^t is appropriate when θ is changing slowly, and thus previous knowledge should be exploited. Exploiting this knowledge will lead to better precision and reliability.

3.2 TEMPORAL FORGETTING FACTOR

Depending on whether the distribution of the parameter θ has changed significantly, a new sample can be drawn according to the *modification* or the *reinforcement* mode. Reinforcement is employed to make the identified probability distribution more precise when it is not undergoing sudden change. Modification is employed otherwise to re-identify the distribution 'from scratch'. Therefore, we define a *model mismatch index*, denoted ζ^t, such that when it surpasses a predefined threshold ($\zeta^t > \zeta_{\text{th}}$), modification is applied. Otherwise ζ^t is used to determine λ^t as follows:

$$\lambda^t = e^{-|\mu_\nu - \zeta^t|}, \tag{9}$$

Algorithm 1 ARMCMC

Assumptions: 1) roughly noise mean (μ_ν) 2) roughly noise variance (σ_ν) 3) desired precision and reliability (ϵ_0, δ_0) 4) desired threshold for model mismatch (ζ_{th})
Goal: Online calculation of parameters posterior distribution given the consecutive t-th pack of data ($P(\theta^t|D^t)$)
Initialization: Prior knowledge for θ_1^0, $n = 0$
Consider desire precision and reliability (ϵ, δ)
repeat
  Put $t_0 = n * N_s + 1$ from (2), $n++$
  Add new data pack to dataset D^t
  Model mismatch index: ζ^t from (10)
  if $\zeta^t < \zeta_{th}$ **then**
    Reinforcement: set prior knowledge equal to the latest posterior of previous pack
    Temporal forgetting factor: λ^t from (9)
  else
    Modification: set prior knowledge θ_1^n
    Temporal forgetting factor: $\lambda^t = 0$
  end if
  Set minimum iteration k_{min} from (12)
  for $k = 1$ **to** k_{min} **do**
    Proposal distribution:
      • draw $\lambda_k \sim U(0, 1)$
      • *Variable jump distribution:* $q_k^t(.)$ from (8)
    Draw $\theta_k^{t*} \sim q_k^t(.)$
    Acceptance rate: $\alpha(.)$ from (6)
    Draw $\gamma \sim U(0, 1)$
    if $\gamma \leq \alpha$ **then**
      'Accept' the proposal
    end if
  end for
  Wait to build $D_{t_0}^{t_{m+N_s+1}}$ (algorithm sample time)
until No data is obtained

where μ_ν is an estimation of the noise mean, by calculation of the expected value in Eq. (1). Note that employing modification is equivalent to setting $\lambda^t = 0$. The model mismatch index ζ^t itself is calculated by averaging the errors of the previous model given the current data:

$$\zeta^t = 1/N_s \sum_{n=1}^{N_s} \Big(y_n^t - \mathop{\mathbb{E}}_{\theta \in \theta^{t-1}} (F(D^t(n), \theta)) \Big), \zeta^0 = \infty \tag{10}$$

Remark 2: The model mismatch index accounts for all sources of uncertainty in the system. To calculate ζ_{th}, one needs to precalculate the persisting error between the predicted and measured data. In other words, ζ_{th} is basically an upper bound for the unmodeled dynamics, disturbances, noises, and any other source of uncertainty in the system.

Remark 3: To avoid numerical issues, the summation of probability logarithms are calculated. In addition, each data pair in the algorithm time sample is weighted based on its temporal distance to the current time. Therefore Eq. (5) is modified as

$$\begin{aligned} \log\big(P(\cdot)\big) &= \sum_{t_m+1}^{t_m+N_s+1} \log P_\nu(e^t), \\ e^t &= \Big(Y_n^t - F^t(D^{t-1}(n), \theta^t) \Big) e^{-\rho(N_s-n)}, \end{aligned} \tag{11}$$

where $\rho \in [0, 1]$ is a design parameter that reflects the volatility of the model parameters, and $e^t = [e_1^t, ..., e_n^t, ..., e_{N_s}^t]$. For systems with fast-paced parameters, ρ should take larger values.

3.3 MINIMUM REQUIRED EVALUATION

Theorem 3.1. *Let ϵ and δ be the desired precision and reliability. Furthermore, assume that the initial sample has enough number of evaluations (as in (7)). To satisfy the inequality in Eq. (7), the minimum number of samples k in ARMCMC is calculated using this implicit relation:*

$$k_{min} = \frac{1}{2\epsilon^2} \log(\frac{2}{\lambda^t(1-\delta) + 2(1-\lambda^t)e^{-2\epsilon^2(1-\lambda^t)k_{min}}}). \tag{12}$$

Proof. Samples from previous pdf: According to the variable jump distribution in (8), given k samples, the expected number of samples drawn from the previous posterior pdf ($P(\theta|D^t)$) is $\lambda^t k$. By assumption, the algorithm has already drawn at least k samples in the previous algorithm time-step. Consequently, by (7), the expected number of samples with distances less than ϵ from $\mathbb{E}(P_k)$ drawn from a previous distribution is at least $\lambda k \delta$.

Samples from Gaussian: By (8), there are $k_0 = (1-\lambda^t)k$ samples drawn in expectation. According to (13), we have $Pr\Big\{\{P_k - \mathbb{E}(P_k)\} \leq \epsilon\Big\} \geq \delta_0$, where δ_0 is given by rearranging (7):

$$\delta_0 = 1 - 2e^{-2\epsilon^2 k_0}. \tag{13}$$

Thus, the expected number of samples with distances less than ϵ from $\mathbb{E}(P_k)$ are at least $\delta_0(1-\lambda^t)k$.

Overall reliability: The total expected number of samples with distances less than ϵ from $\mathbb{E}(P_k)$ is the summation of the two parts mentioned above. Hence it is obtained through dividing by k:

$$\delta_1 = \frac{(\lambda^t k\delta) + (\delta_0(1-\lambda^t)k)}{k} \tag{14}$$

Given the new obtained reliability, which is greater than the desired one, helps us decrease the number of evaluations. For the sake of illustration, Fig. 2 presents the minimum required number of evaluations with respect to λ for different precisions and reliabilities. As it can be seen, the MCMC is equal to ARMCM if λ is always set to one. The number of evaluations in ARMCMC mitigates as the validity of the previous model increases. □

4 RESULTS

In this section, we demonstrate the performance of the proposed approach on two examples. First, we employ the proposed method to identify parameters in the soft bending actuator model and compare the results with identification using Recursive Least Squares (RLS). In the second example, we evaluate our method on the Hunt-Crossley model given reality-based model and compare it with simple MCMC and RLS. All the results/code are available on GitHub.

4.1 FLUID SOFT BENDING ACTUATOR

Consider the dynamic model of a fluid soft bending actuator, given by Wang et al. (2019):

$$\begin{aligned} \ddot{\alpha} &= q_1(p - p_{atm}) - q_2\dot{\alpha} - q_3\alpha, \\ u_c \,\mathrm{sign}(p_s - p)\sqrt{|p_s - p|} &= q_4\dot{p} + q_5\dot{p}p, u_d = 0, \\ u_d \,\mathrm{sign}(p - p_{atm})\sqrt{|p - p_{atm}|} &= q_6\dot{p} + q_7\dot{p}p, u_c = 0, \end{aligned} \tag{15}$$

where α is the angle of the actuator, u_c, u_d are the control inputs for retraction and contraction, respectively. Also p, p_s, p_{atm} are the current, compressor and atmosphere pressure respectively. For this example, we assume $q_1 = 1408.50, q_2 = 132.28, q_3 = 3319.40$ are known and $p_{atm} = 101.3$ kPa, $p_s = 800$ kPa. We are trying to identify the four other parameters $(q_4, ..., q_7)$. The true parameters are $q_5 = -2.14 \times 10^{-4}, q_6 = 6.12 \times 10^{-9}, q_7 = -9.76 \times 10^{-5}, q_8 = -1.90 \times 10^{-9}$. To this end, we assume the hybrid model below:

$$u \;\mathrm{sign}(\Delta p)\sqrt{|\Delta p|} = \theta_1\dot{p} + \theta_2\dot{p}p, u = \{u_c, u_d\}, \tag{16}$$

where Δp is either $(p_s - p)$ for retraction or $(p - p_{atm})$ for contraction. As the range of parameters are small, we scale the input vector by the factor of 10^7 for RLS. Given the input (u_c, u_d) and the

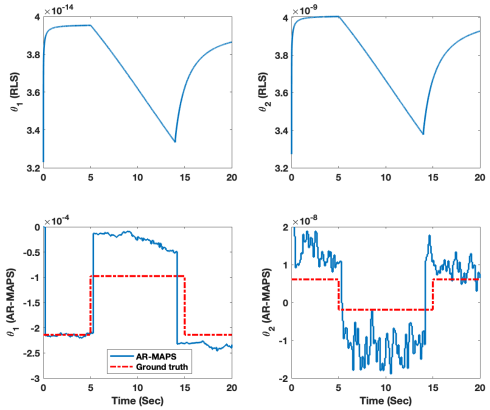


Figure 3: Parameter variation for RLS and AR-MAPS

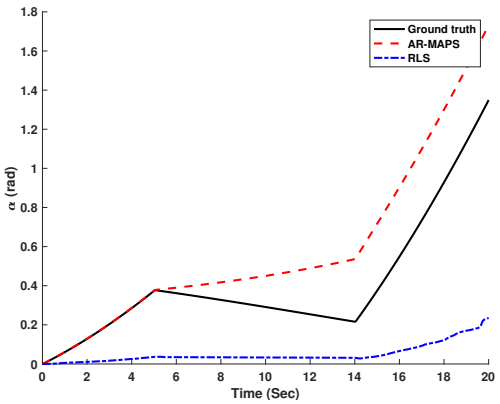


Figure 4: Angle of the actuator comparison of RLS and AR-MAPS for soft bending actuator.

output $(p, \dot{p})$, we want to identify the parameter and estimate the current angle of actuator assuming that its initial position at the origin. The data sample time is $T = 1$ ms and each data pack includes 100 samples which results in an algorithm sample time equal to $T_s = 0.1$ sec. Point estimation obtained by considering the mode at the modification phase and the median during the reinforcement phase; this estimate is denoted as AR-MAPS. The point estimate results for the parameter estimation are shown in Fig. 3. The estimation errors are $0.0235, 6.0053 \times 10^{-7}$ for θ_1, θ_2 in RLS and $0.0089, 1.1840 \times 10^{-7}$ in AR-MAPS, respectively. Moreover, the estimation of the angle is plotted in Fig. 4.

4.2 Hunt-Crossley model

In this section, we demonstrate ARMCMC by identifying parameters of the Hunt-Crossley model, which represents an environment involving a needle contacting soft material. The needle is mounted as an end-effector on a high-precision robotic arm, which switches between two modes: free motion and contact. Due to abrupt changes in the model parameters when the contact is established or lost, online estimation of the force is extremely challenging.

4.2.1 Contact dynamic model

Consider the dynamics of contact as described by the Hunt-Crossley model, which is more consistent with the physics of contact than classical linear models such as Kelvin-Voigt (Haddadi & Hashtrudi-Zaad, 2012). In order to overcome the shortcomings of linear models, Hunt & Crossley (1975) proposed the following hybrid nonlinear model:

$$f_e(x(t)) = \begin{cases} K_e x^p(t) + B_e x^p(t)\dot{x}(t) & x(t) \geq 0 \\ 0 & x(t) < 0, \end{cases} \tag{17}$$

in which $K_e, B_e x^p$ denote the nonlinear elastic and viscous force coefficients, respectively. The parameter p is typically between 1 and 2, depending on the material and the geometric properties of contact. Also, $x(t), \dot{x}(t), f_e$ are the current position, velocity (as input X) and contact force (as output (Y in Eq. (1)) of a needle near or inside the soft material, with $x \geq 0$ representing the needle being inside. This needle can move freely in open space or penetrate the soft contact; the forces on this needle are modeled using the Hunt-Crossley model. The practical problem we consider is to estimate the force at the tip of the needle by identifying the model parameters. K_e, B_e, p are three unknown parameters (θ in Eq. (1)) that needs to be estimated. An online estimate of contact force plays a pivotal role in the stable interaction between robotic manipulators and unknown environments.

$$\begin{aligned} \log(f_e) &= \log(K_e x_s^p + B_e \dot{x}_s x_s^p), \\ \log(f_e) &= p \log(x_s) + \log(K_e + B_e \dot{x}_s). \end{aligned} \tag{18}$$

For RLS, we also need to make the assumption that $B_e/K_e\dot{x}_s << 1$. Note that the vector of parameters (θ) in the following relation are not independent, which may lead to divergence. With this assumption, we have

$$\begin{aligned} \log(1 + B_e/K_e\dot{x}_s) &\approx B_e/K_e\dot{x}_s, \\ \log(f_e) &= p\log(x_s) + \log(K_e) + B_e/K_e\dot{x}_s. \end{aligned} \tag{19}$$

$$\begin{aligned} \phi &= [1, \dot{x}_s, \log(x_s)], \\ \theta &= [\log(K_e), B_e/K_e, p]^T. \end{aligned} \tag{20}$$

4.2.2 SETUP

The data structure is same as previous simulation. Prior distribution of all three parameters (K_e, B_e, p) are initialized to $N(1, 0.1)$ (a normal distribution with $\mu = 1$ and $\sigma = 0.1$) Moreover, as more data is collected, the spread of the posterior pdf decreases. A bit after 5 seconds, the needle goes outside of the soft material, and experiences zero force; this is equivalent to all parameters being set to zero. The color-based visualization of probability distribution over time is used for the three parameters in Fig. 5. During the period of time that the whole space is blue (zero probability density), there is no contact and the parameter values are equal to zero.

Since we are taking a Bayesian approach, we are able to estimate the entire posterior pdf. However, for the sake of illustration, the point estimates are computed from the ARMCMC algorithm by using AR-MAPS method. The results are shown in Fig. 6 for the time-varying parameters $\theta_1 = K_e, \theta_2 = B_e, \theta_3 = p$. During the times that RLS results are chattering due to the use of saturation (if not, the results would have diverged), the needle is transitioning from being inside the soft material to the outside or vice versa. In addition, due to the assumption (19), performance can deteriorate even when there is no mode transition. Furthermore, in the RLS approach, estimated parameters suddenly diverged during free motion, since the regression vectors are linearly dependent. In contrast, with ARMCMC this is not an issue we need to consider. The result of ARMCMC is presented in Fig. 7, which shows the force estimation with two different identification approaches. This probability of interest can be easily obtained by deriving the parameter density at one's disposal.

4.2.3 QUANTITATIVE COMPARISON

Quantitative details of comparing a naive point estimate of the ARMCMC algorithm by averaging the particles (denoted as AR-APS) and the RLS method are listed in Table 1. This reveals more than a 70% improvement in the precision of all model parameters throughout the study by using the Mean Absolute Error (MAE) criteria and also more than a 55% improvements in the force estimation error. Among parameters, the viscose (B_e) has the largest error in the RLS method since it is underestimated due to the restrictive assumption in Eq. (19). The AR-MAPS approach uplifts the performance of the parameter identification and the force estimation.

We also compare ARMCMC to MCMC. For the algorithm to run in real-time, MCMC requires more time to converge. For this example, with $\lambda = 0.7$, the value of k_{min} is 15000 for MCMC but only 6000 for ARMCMC with $\epsilon = 0.01, \delta = 0.9$. Two possible ways to address this drawback in MCMC are to reduce the number of samples to 5000 per algorithm iteration (denoted MCMC-1 in Table 1), which results in worse precision and reliability compared to ARMCMC, or to increase the algorithm sample time to 0.2 (denoted MCMC-2 in Table 1) which would cause more delay in the estimation result and slower responses to changes in the parameter.

5 CONCLUSIONS

This paper presented an adaptive recursive MCMC (ARMCMC), algorithm for an online identification of full probability distribution of model parameters in a Bayesian paradigm. When applied to systems involving abrupt changes of model parameters which can occur when contact with a soft environment is established or lost, conventional approaches suffer from low performance. Empirical results on the Hunt-Crossley model as a nonlinear hybrid dynamic model was compared with a well-known conventional identification approaches and revealed proficiency of the proposed algorithm. The proposed method adapts quickly to abrupt changes which relaxes the pre-requirement

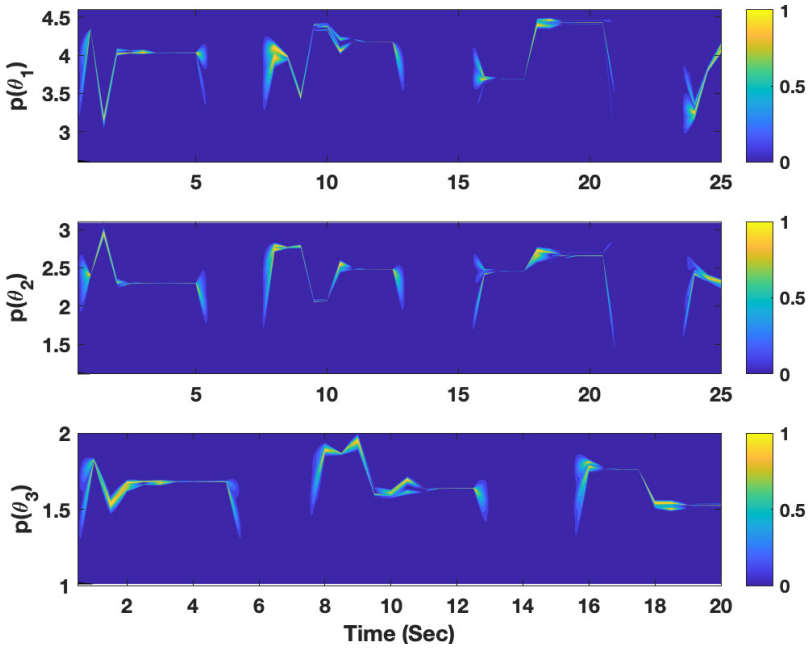


Figure 5: Probability distribution of parameters ($\theta_1 = K_e, \theta_2 = B_e, \theta_3 = p$) using ARMCMC.

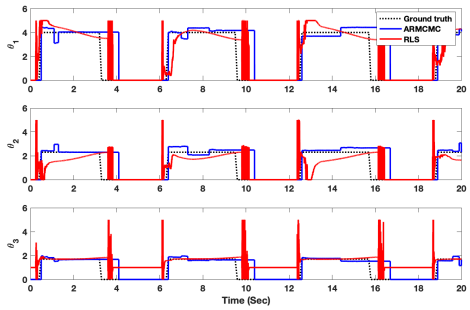


Figure 6: Model parameters ($\theta_1 = K_e, \theta_2 = B_e, \theta_3 = p$) point estimation in AR-MAPS.

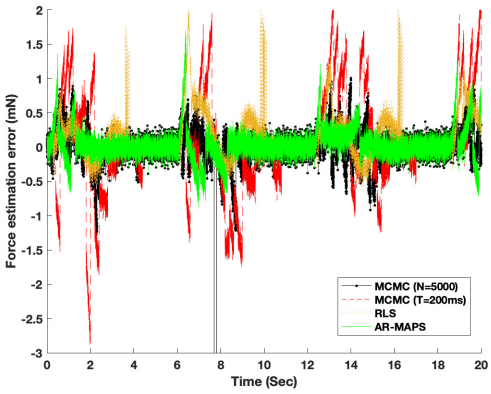


Figure 7: Force prediction error in RLS, AR-APS, and MCMC.

Table 1: Comparison of RLS (Haddadi & Hashtrudi-Zaad, 2012) and point estimate of ARMCMC and MCMC for environment identification.

ERRORS (MAE)	K_e	B_e	p	F_e (MN)
RLS	0.5793	0.9642	0.3124	51.745
MCMC-1	0.6846	0.8392	0.3783	76.695
MCMC-2	0.7294	0.9964	0.4195	101.88
AR-APS	0.0774	0.0347	0.0945	33.774
AR-MAPS	0.0617	0.0316	0.0756	31.659

conditions in the parameters. As future work, deploying a fully probabilistic framework from identification to control and a decision-making stage is considered to exploit the full potentials of the Bayesian optimization. Additionally, employing a method to compensate the delay will be taken into consideration.

REFERENCES

Niki Abolhassani, Rajni Patel, and Mehrdad Moallem. Needle insertion into soft tissue: A survey. *Medical Engineering and Physics*, 29(4):413–431, 2007.

Pedram Agand and Mahdi Aliyari Shoorehdeli. Adaptive model learning of neural networks with uub stability for robot dynamic estimation. In *2019 International Joint Conference on Neural Networks (IJCNN)*, pp. 1–6. IEEE, 2019.

Pedram Agand, Hamid D Taghirad, and Ali Khaki-Sedigh. Particle filters for non-gaussian hunt-crossley model of environment in bilateral teleoperation. In *4th International Conference on Robotics and Mechatronics (ICROM)*, pp. 512–517. IEEE, 2016.

S Bhasin, K Dupree, PM Patre, and WE Dixon. Neural network control of a robot interacting with an uncertain hunt-crossley viscoelastic environment. In *Dynamic Systems and Control Conference*, volume 43352, pp. 875–882, 2008.

Christopher M Bishop. Pattern recognition. *Machine Learning*, 128, 2006.

Steve Brooks, Andrew Gelman, Galin Jones, and Xiao-Li Meng. *Handbook of markov chain monte carlo*. CRC press, 2011.

André S Carvalho and Jorge M Martins. Exact restitution and generalizations for the hunt–crossley contact model. *Mechanism and Machine Theory*, 139:174–194, 2019.

Noah J Cowan, Ken Goldberg, Gregory S Chirikjian, Gabor Fichtinger, Ron Alterovitz, Kyle B Reed, Vinutha Kallem, Wooram Park, Sarthak Misra, and Allison M Okamura. Robotic needle steering: Design, modeling, planning, and image guidance. In *Surgical Robotics*, pp. 557–582. Springer, 2011.

Nicola Diolaiti, Claudio Melchiorri, and Stefano Stramigioli. Contact impedance estimation for robotic systems. *IEEE Transactions on Robotics*, 21(5):925–935, 2005.

Peter J Green. Reversible jump markov chain monte carlo computation and bayesian model determination. *Biometrika*, 82(4):711–732, 1995.

Peter L Green. Bayesian system identification of a nonlinear dynamical system using a novel variant of simulated annealing. *Mechanical Systems and Signal Processing*, 52:133–146, 2015.

Amir Haddadi and Keyvan Hashtrudi-Zaad. Real-time identification of hunt-crossley dynamic models of contact environments. *IEEE transactions on robotics*, 28(3):555–566, 2012.

KH Hunt and FRE Crossley. Coefficient of restitution interpreted as damping in vibroimpact. *Journal of applied mechanics*, 42(2):440–445, 1975.

Dominik Joho, Gian Diego Tipaldi, Nikolas Engelhard, Cyrill Stachniss, and Wolfram Burgard. Nonparametric bayesian models for unsupervised scene analysis and reconstruction. *Robotics: Science and Systems VIII*, pp. 161, 2013.

Shima Khatibisepehr, Biao Huang, and Swanand Khare. Design of inferential sensors in the process industry: A review of bayesian methods. *Journal of Process Control*, 23(10):1575–1596, 2013.

Scott Kuindersma, Roderic Grupen, and Andrew Barto. Variational bayesian optimization for runtime risk-sensitive control. *Robotics: Science and Systems VIII*, 2012.

Tomasz Kuśmierczyk, Joseph Sakaya, and Arto Klami. Variational bayesian decision-making for continuous utilities. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d Alché-Buc, E. Fox, and R. Garnett (eds.), *Advances in Neural Information Processing Systems 32*, pp. 6392–6402. Curran Associates, Inc., 2019.

Brett Ninness and Soren Henriksen. Bayesian system identification via markov chain monte carlo techniques. *Automatica*, 46(1):40–51, 2010.

Ali Noormohammadi-Asl and Hamid D Taghirad. Multi-goal motion planning using traveling salesman problem in belief space. *Information Sciences*, 471:164–184, 2019.

Kaur Aare Saar, Fabio Giardina, and Fumiya Iida. Model-free design optimization of a hopping robot and its comparison with a human designer. *IEEE Robotics and Automation Letters*, 3(2): 1245–1251, 2018.

Roberto Tempo, Giuseppe Calafiore, and Fabrizio Dabbene. *Randomized algorithms for analysis and control of uncertain systems: with applications*. Springer Science & Business Media, 2012.

Felipe Tobar. Bayesian nonparametric spectral estimation. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett (eds.), *Advances in Neural Information Processing Systems 31*, pp. 10127–10137. Curran Associates, Inc., 2018.

Beilun Wang, Arshdeep Sekhon, and Yanjun Qi. A fast and scalable joint estimator for integrating additional knowledge in learning multiple related sparse Gaussian graphical models. In Jennifer Dy and Andreas Krause (eds.), *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pp. 5161–5170, Stockholmsmässan, Stockholm Sweden, 10–15 Jul 2018. PMLR.

Tao Wang, Yunce Zhang, Zheng Chen, and Shiqiang Zhu. Parameter identification and model-based nonlinear robust control of fluidic soft bending actuators. *IEEE/ASME Transactions on Mechatronics*, 24(3):1346–1355, 2019.

A Appendix

According to Abolhassani et al. (2007), a nonlinear hybrid model based on a reality-based soft environment is considered as follows:

$$f_e = f_{\text{st}}(x, t, t_p) + f_{\text{fr}}(x, \dot{x}) + f_{\text{ct}}(x, t, t_p), \tag{21}$$

where x is the needle tip position and t_p is the latest time of puncture. Initial position of the environment is assumed to be at the origin. The stiffness of the force (f_{st}) belongs to a pre-puncture and the friction (f_{fr}) and cutting forces (f_{ct}) belong to a post-puncture. The stiffness force is modeled using a nonlinear Hunt-Crossley model:

$$f_{st}(x, t, t_p) = \begin{cases} 0 & x < 0 \\ K_e x^p(t) & 0 \leq x \leq x_1, t < t_p \\ 0 & x > x_2, t \geq t_p \end{cases} \tag{22}$$

where K_e, p are the same parameters defined in (17). The maximum depth that the soft environment yields before the puncture and its position after it is denoted by x_1, x_2, respectively ($0 < x_2 < x_1$). In this study, the needle can insert up to 16.65, 10.21 mm before and after penetration. A friction model is inspired from modified Karnopp model.

$$f_{fr}(x, \dot{x}) = \begin{cases} C_n sgn(\dot{x}) + B_e x^p \dot{x} & \dot{x} \leq -\Delta v/2 \\ \max(D_n, F_a) & -\Delta v/2 < \dot{x} \leq 0 \\ \max(D_p, F_a) & 0 < \dot{x} < \Delta v/2 \\ C_p sgn(\dot{x}) + B_e x^p \dot{x} & \dot{x} \geq \Delta v/2 \end{cases} \tag{23}$$

where $C_n = -11.96 \times 10^{-3}$ and $C_p = 10.57 \times 10^{-3}$ are negative and positive values of dynamic friction, $D_n = -0.01823$ and $D_p = 0.01845$ are negative and positive values of static friction, and B_e, p are same as Eq. (17). The relative velocity between the needle and tissue is denoted by $\dot{x}$, and $\Delta v/2 = 0.005$ is the value below where the velocity is considered to be zero, and F_a is the sum of non-frictional applied forces. The cutting force is considered as a static force constant for the tissue and the needle geometry if soft tissues are not inhomogeneous and anisotropic (Cowan et al., 2011)

$$f_{ct}(x, t, t_p) = \begin{cases} 0 & x \leq x_1, t < t_p \\ 0.94 & x > x_2, t \geq t_p \end{cases}. \tag{24}$$

According to the previous relations, the system is considered as a hybrid model while providing both free motion and in-contact environment. The manipulator is a translational mechanism with a friction, slip, and hysteresis loop for the actuator. To present the superiority of the proposed algorithm, the results are compared with the RLS method presented by Haddadi & Hashtrudi-Zaad (2012). To prevent the results of RLS from divergence in model mismatch sequences, saturation is applied in the outputs of the identifier.